

IS1128 IT, Management, Organization

Module focuses on information technologies (IT) in organizations and the interplay between IT, work, management, and organizations.

- Multifaceted roles of IT, support communication, collaboration, and organizational improvements in operations, planning, and decision making.
- Opportunities for IT-enabled innovations and issues involving information systems (IS) adoption and deployment.
- Impacts of modern IT and related artificial intelligence (AI) technologies on knowledge workers, teamwork, work design, management practices, and organization.

1. Impact of IT/IS on Businesses

Global spending on information technology (IT) and IT services steadily increasing. Organizational, management, and cultural changes required for firms to derive full business value from IT investment. But does not guarantee good returns. Overall impacts on organization (structure, culture, team capability), technology, management.

Strategic business objectives

- **Survival:** Information technologies as necessity of business. *Innovation Diffusion Model:* Roger's bell curve adoption of technology (Innovators, early adopters, early/late majority, laggards).
- **Operational excellence:** Improvement of efficiency and effectiveness to attain higher profitability. (e.g. accounting systems, marketing forecasters, production management software). *Automating (faster), Learning (better - exploitation and exploration), Strategizing (smarter than rivals)*. E.g. Walmart VMI Retail Link reduces bullwhip effect.
- **New products/service/business model:** E.g. sharing economy, Netflix.
- **Customer and supplier intimacy:** Customers served well become repeat customers. Close relationships with suppliers result in lower costs. E.g. investment in CRM solutions.
- **Improved Decision Making:** Real-time, accurate information. Better forecasts, planning, response time. E.g. Jooz real time metrics, Starbucks GIS.
- **Competitive advantage:** Advantage over competitors. Better performance, price competitive. Sometimes, create brand image of innovation at initial loss.

Overall, growing interdependence between information technology and implementing corporate strategies to achieve corporate goal. We assume a **sociotechnical view**, where optimal organizational performance achieved by jointly optimizing both social and technical systems used in production. Helps avoid purely technological approach.

2. IS and Organizations

Features of organizations relevant to build, use IT/IS successfully: Org. structure, Business model, Business process, Business/IT alignment.

Organization

- **Technical:** A stable, formal social structure that takes resources from the environment and processes them to produce inputs. Formal legal entities with internal rules, procedures.
- **Behavioral:** A collection of rights, privileges, obligations, and responsibilities delicately balanced through conflict and conflict resolution.
- Organizations are Bureaucracies with clear-cut divisions of labor and specialization, adhere to *principle of efficiency* to maximizing output using limited inputs.

Organizational Structure

The formal configuration between individuals and groups regarding the allocation of tasks, responsibilities, and authority within the organization. We define 5 basic kinds:

- **Entrepreneurial:** Simple structure, Fast-changing environment, Major decision made by one or two key personnel, Direct supervision, e.g., small start-up business.
- **Machine Bureaucracy:** Centralized management/decision making, Stable environment, High degree of formalization and work specialization, Standardization of work processes, Many levels in the chain of command, Goal: achieve internal efficiency, e.g., midsize manufacturing firm, government agencies.
- **Divisionalized Bureaucracy:** Multiple machine bureaucracies, Each producing a different product or service, Decentralized decision-making at the divisional level, Little coordination among separate divisions, Topped by a central headquarters, e.g., Fortune 500 firms with multiple product lines.
- **Professional Bureaucracy:** Knowledge-based organization, Standardization of skills, Goods and services depend on professionals' expertise, Highly trained professionals provide nonroutine services, Formalized but decentralized to give autonomy, Department heads with weak centralized authority, Small top management, Few middle managers, Goal: Innovate and provide high-quality services, e.g., law firms, universities, hospitals.
- **Adhocracy:** Low formalization and decentralization, Task-force organization responding to a rapidly changing environment, Engages in nonroutine tasks and uses sophisticated technology, Goal: Innovation and rapid adaptation, Large groups of specialists in short-lived multidisciplinary teams, Weak central management, e.g., NASA, Pixar Animation Studios.

Information systems often reflect organizational structure, such as centralization or decentralization.

Business Model

A business model specifies how a company will create, deliver, and capture value. What does company do? How unique? Revenue, Cost, key resources / activities / partners needed? Business Model Canvas.

- **Business Processes:** Logically related set of activities/steps that define how specific business tasks are performed. Flows of material, information, knowledge, tied to functional area or be cross-functional
- Businesses can be seen as collection of business processes, business processes may be assets or liabilities.
- Focus on alignment of IT strategy with business's strategy. Continuous process of adjusting business goals and IS architecture to achieve business objectives, support/enable business transformation.

3. IS and Competition Strategies

Focus on how Porter's competitive forces model, the value chain model, and SWOT analysis help companies develop competitive advantages (CA) using IT/IS. Competition strategies for organizations to handle disruptive technology and disruptions.

A. Where to Compete: Analyzing Competitive Forces 5 Forces

Analyze competitive forces to focus resources effectively. Michael Porter's Five Forces Model provides a general view of a firm, competitors, and environment.

- **Rivalry among existing competitors:** High intensity if competitors are numerous or roughly equal in size and power. Slow Industry growth, High exit barriers. Rivals are highly committed with leadership aspirations. Firms struggle to read each other's signals due to unfamiliarity.
- **Threat of new entrants:** Barriers to entry include Supply-side economies of scale, Demand-side benefits of scale. Customer switching costs. Capital requirements. Incumbency advantages independent of size. Unequal access to distribution channels. Restrictive government policies.
- **Bargaining power of suppliers** Suppliers are powerful if: More concentrated than the industry they sell to. Do not depend heavily on the industry for revenue. Switching costs in changing suppliers are high. Products differentiated. No substitutes. Can credibly threaten forward integration.
- **Bargaining power of buyers** Buyers have strong negotiation power if: few buyers or large purchase volume. Products standardized or undifferentiated. Switching costs low. Can credibly threaten backward integration.
- **Threat of substitute products or services** High threat if substitutes offer attractive price-performance trade-off and switching costs low.

Type of Business

- **Pipeline Business:** Creates value by controlling a linear series of activities. Follows the classic value chain model. Inputs (e.g., materials from suppliers) undergo transformations into a finished product with higher value. - E.g. Nike, Dell.
- **Platform Business** Facilitates interactions between producers and consumers. *Critical asset:* Community. Resources of participating groups. Strategic focus shifts from: Resource control to orchestrating resources, Internal process optimization to facilitating external interactions, End-user value to maximizing ecosystem value. - E.g. Gojek, Airbnb.

Network Effects

- **Direct Network Effects (One-Side Effects):** Utility user receives increases as the number of users grows. E.g. Communication platforms. *Differentiation:* Capacity-constrained vs. capacity-unconstrained assets.
- **Indirect Network Effects (Cross-Side Effects):** Value of a service for one user group rises when another user group expands. Critical for multi-sided platforms.

Market Concentration

- **Concentration Ratio (CR):** Measures the combined market share percentage of the largest firms in an industry. - Typically calculated for the top three or four firms (e.g., CR4). Market types: Monopoly: CR1 100%, Oligopoly: CR5 > 60%, Pure competition: CR4 0%.
- **Herfindahl-Hirschman Index (HHI):** Derived by summing the squares of all firms' market shares. Higher values equal dominance by large players. Value ranges: Decimal format: (0,1]. Whole number format: (0,10000].
- Sample reference: HHI < 1500 non-concentrated, HHI 1500-2500 moderate, HHI > 2500 is highly concentrated.

B. How to Compete: Choosing Generic Strategy

- **Low cost leadership strategy:** Lower price than competitors, greater efficiency. Keep tight control of overheads and maximizing the cost benefit of industrial experience and new technology. Avoid unprofitable/marginal customer accounts, minimize running cost or investment in processes seen as ancillary, such as R&D, advertisement, and customer service.
- **Product differentiation** New products/services, Innovation, high start-up and running cost.

- **Market Niche** Specific market focus.

C. Identify Resources and Capabilities

- **Resources:** Organization's specific assets utilizable to achieve cost or product differentiation E.g., proprietary technology, brand equity, a loyal and established customer base
- **Capability:** The organization's ability to leverage these resources in the marketplace.
- **Distinctive Competencies:** innovation, agility, quality, or low cost. The resources and capabilities develop these competencies, that contribute to some sustained competitive advantages.

SWOT Analysis

Strengths, Weaknesses, Opportunities, and Threats. SWOT analysis is a framework to help assess and understand the internal and external forces that may create opportunities or risks for an organization.

- **Strengths:** Traits giving advantage over competitors
- **Weaknesses:** Traits specific to the business that put it at disadvantage.
- **Opportunities:** Elements in external environment that allow formulation, implementation of strategies.
- **Threats:** Threats in external environment that could endanger business, CA, profits. (Not specific)

D. Identify How to Compete: Analyzing the Value Chain

- Organizations as series of activities adding value to products or services (value chain). Analysis helps determine where value is added and the costs incurred. Highlights areas where competitive strategies can be applied.
- Differentiates between:
 - **Primary activities** (e.g., inbound logistics, operations, marketing, sales, service).
 - **Support activities** (e.g., firm infrastructure, HR, technology development, procurement).
- IS can improve efficiency, strengthen customer/supplier relationships. E.g. through benchmarking, industry best practices.

Disruptive Technologies

Technologies that introduce sweeping changes to industries and markets by outperforming existing products. Provides substitute products that perform as well or better, and drives transformation across industries. E.g. Internet, IoT, Mobile payments, Generative AI (GenAI). Need to adapt to digital disruptions.

Adaptation Strategies

- **Fight Back:** Compete directly through new units, acquisitions, or joint ventures. E.g. Carmakers shifting towards electric vehicles.
- **Double Down:** Strengthen existing advantages and focus on core business. (Think it is a fad, ignore, low ROI)
- **Retrench:** Yield ground to new entrants and conduct survival tactics. E.g. Mergers, acquisitions, lobbying.
- **Move Away:** Shift towards new business opportunities.

Two Strategic Dimensions for these adaptation strategies: Existing Market vs. New Market and Offensive Strategy vs. Defensive Strategy. Companies always end up doing some hybrid.

- **Successful Adaptations:** Business innovation, quick to adapt.

- **Failed Adaptations:** Failure to embrace new business models enabled by digital disruption. Business myopia focus too much on existing revenue streams.
- **Considerations:** What is the value proposition? What new opportunities does disruption open up (business/tech trend)? What capabilities needed to realize opportunities?

4. IS Strategic Deployment

Factors organizations need to consider in IT planning and implementation. Includes system acquisition strategies, system development/delivery methods, system conversion strategies.

- **Internally:** Consider perspective of Business/IT Alignment, Organizational, managerial IT features.
- **Externally:** PESTLE analysis. (Political, Economic, Sociological, Technological, Legal, Environment). E.g. govt regulations, inflation rate, religion, laws, env. concerns.

Feasibility Analysis

- **Organizational Feasibility:** how well solution fits within the organization's strategic planning objectives and goals.
- **Political Feasibility:** how well solution fits within the organization's political environment.
- **Technological Feasibility:** how well solution supportable given organization's current technology infrastructure and resources, (hardware, software, networking, personnel).
- **Behavioral/Operational Feasibility:** how well fits within the organization's culture and the extent to which users are expected to accept the solution. (E.g. Amazon surveillance of workers)
- **Economic Feasibility:** net economic benefits of a particular solution: Cost-Benefit Analysis (CBA)

System Acquisition Strategies

- **Internal Acquisition:** In-house development. Optimal if we need full control of the system. *Challenges include* limited IT staff, IT staff having a limited skill set, overworked IT staff, and problems with performance of IT staff.
- **External Acquisition:** External acquisition of a prepackaged system. Commercial off-the-shelf software package (E.g. Cloud solutions). Outsourcing system development for Customization.
- **Acquisition Strategies:** Important to carefully consider the acquisition strategy. *Strategic tasks* enhance the organization's competitiveness, while critical tasks support core and essential business operations. Routine tasks should also be evaluated.
- Key factors in acquisition decisions include *strategic importance*, cost, quality, and regulation.

External Acquisition Steps

1. **Systems Planning and Selection:** Identify business needs and determine whether an external acquisition is necessary.
2. **Systems Analysis:** Define system requirements, analyze existing workflows, and establish key functionalities needed.

3. **Development of a Request For Proposal (RFP)** – document outlining requirements for vendors, may include: summary of existing systems and applications, requirements for system performance and features, reliability, backup, and service requirements, criteria that will be used to evaluate proposals, timetable and budget constraints)
4. **Proposal Evaluation:** Assess vendor responses based on predefined criteria, including technical capabilities, cost, and support services.
5. **Vendor Selection:** Choose the most suitable vendor based on evaluation outcomes and proceed with contract negotiation and implementation.

Cloud Solutions: Public cloud (third-part service providers), Private cloud (operated solely for organization, or hosted internally), Hybrid cloud (combination).

Internal Acquisition Methods

System Development Methods: Organizations can adopt various approaches to internal system development, each with its own advantages and challenges.

- **Traditional Software Development Lifecycle (SDLC):** A phased "waterfall" approach that follows sequential stages—planning, analysis, design, implementation, and maintenance. While structured, it can be costly, time-consuming, and inflexible.
 - **Agile Methodology:** Focuses on iterative development with quick releases of working software. Encourages adaptability to changing requirements and continuous collaboration between stakeholders.
 - **CI/CD (Continuous Integration and Continuous Deployment/Delivery):** Automates code integration, testing, and deployment, allowing frequent and reliable software updates while minimizing manual errors.
 - **Lean Startup:** Emphasizes experimentation over detailed planning, customer feedback over intuition, and iterative design over traditional long-term development cycles.
 - **MVP (Minimum Viable Product):** A strategy where a product is developed with just enough features to attract early adopters and validate the idea before investing in full-scale development.
 - **Prototyping:** A trial-and-error approach used to explore system functionality. It is particularly useful when requirements are unclear or when iterative feedback from stakeholders is necessary.
 - **DevOps:** A collaborative approach that integrates software development and IT operations. It follows agile principles and promotes automation, continuous monitoring, and rapid deployment to enhance system stability and performance.
- ### System Conversion Strategies
- **Conversion:** Process of changing from old system to new system. Four main strategies
 - **Parallel strategy:** Safest, but expensive
 - **Direct cutover strategy:** Risky, but fast
 - **Pilot study strategy:** Introduces the system to only a limited area of the organization first
 - **Phased approach strategy:** Introduce the system in stage, either by functions or by organization units
 - Requires end-user training. Finalization of detailed documentation showing how the system works from both technical and end-user standpoint

5. IT Project and Change Management

Project management strategies and practices at organizational and project level. Change management challenges, pitfalls, strategies and tactics.

IT Project Portfolio Analysis

- Inventories all of the organization's information systems projects and assets.
- Strategic goals & objectives, Classification of project types:
 - **Breakthroughs:** E.g., Digital twins in factory.
 - **Incremental:** Enterprise search proj (internal productivity system).
 - **Mandatory:** Cybersecurity.
- Budget allocation and commitment to specific projects: Naturally, greater the value to business, and lower the risk, the higher the funding priority. Otherwise, fund selectively or do not fund. (Project categorization affects funding decision).
- Ultimately balance risk and return from systems investments to improve return on portfolio.

Discussion: Unilever's Digital Transformation

- **Digital Twins:** A virtual representation of an object or system that spans its lifecycle, updated from real-time data, using simulation, machine learning, and reasoning to aid decision-making.
- Unilever's digital twin solution was custom-built by their engineering team in partnership with The Marsden Group (a Microsoft partner) and hosted on Microsoft's Azure platform. However, as risk is high (high value, high risk) - currently, Unilever experiments on eight digital twins first across NA, SA, Europe, Asia, streaming data from 15 of its 300 global plants.

Factors in IT Project Portfolio Analysis

- Standardization: Applications and infrastructure are well-defined and documented.
- Centralization and demand management (Well-defined scheme for screening, categorizing, and prioritizing projects).
- Metrics in project prioritization: Strategic alignment, benefits measurement, feedback mechanisms, and advanced valuations (e.g., inclusion of qualitative option value in funding decisions).

Tools

- **Competing Investments: Weighted Multicriteria Analysis:** Used to evaluate alternatives when multiple criteria exist. Assigns weights to various features of a system and calculates weighted total scores.
- **Making the Business Case:** three types of arguments commonly made in the business case for an IS:
 - **Faith:** Based on beliefs about organizational strategy, competitive advantage, industry forces, etc.
 - **Fear:** Based on the notion that failure to implement the system will result in loss to competition or business failure.
 - **Facts:** Based on data, quantitative analysis, and indisputable factors.

- **Cost-Benefit Analysis and Total Cost of Ownership (TCO)** TCO includes direct and indirect costs (capital vs. operational costs) Includes things like training, support, maintenance costs. Capital cost usually very low (maybe 20%?).

- Benefits can be split into tangible and intangible. Tangible benefits (cost saving): Increased productivity, lower operational costs, reduced workforce, etc. Intangible benefits: Improved asset utilization, enhanced customer experience, increased job satisfaction, etc.

The Productivity Paradox

- While costs of developing an IS are easy to quantify, tangible productivity gains are often difficult to measure.
- Challenges include measurement problems (measure effectiveness and efficiency), time lags (between investment and obsv. impacts), redistribution of resources (industry level productivity is the same - zero-sum game, if rival company up, others lose market share), and mismanagement (not utilizing IS properly).

Project Management Objectives

- A project is a planned series of related activities for achieving a specific business objective. IT/IS project management deals with five major variables: **Scope, Time, Cost, Quality, Risk..** Time hardest to meet. Classic trilemma (2 out of 3) - time, cost, quality. Factors influencing project success: Project size, project structure, experience with technology.

Planning Tools

- **Gantt Chart:** Visually represents tasks, timelines, and resource requirements. Each task as a horizontal bar whose length is proportional to the time required to complete it.
- **PERT Chart:** Graphically depicts project tasks and their interrelationships.
- **Agile Approach:** Iterative development with continuous feedback and adaptation.

Change Management

- A systematic approach to managing change at both organizational and individual levels. Activities include communication, training, and process change.
- Role of a **change agent:** Redefines configurations, interactions, job activities, and power relationships. (can be internal or external agent, no best way. E.g. NUS LMS transition - internal)
- **Sources of resistance:** Organizational level: Structural inertia, threat to power relationships, resource allocation. Individual level: Habit, fear of the unknown, economic factors.
- **Models of adoption of innovation:** Demand-driven - if benefits higher than costs, adopt. Performance, effort expectancy, Social influence, Facilitating condition. Supply-driven - resistance to innovation due to knowledge barriers. Remove barriers helps to diffuse innovation. Organizational learning, skills development.

Typical Pitfalls in Change Management

- Underestimating the magnitude of change (structural and cultural).
- Poor leadership and lack of commitment to change.
- Treating change management as a "nice to have" rather than a critical success factor. (Failing to embed change management steps in the project plan.)

Change Management Strategies and Tactics

- Define clear expectations and roles; include key managers and personnel.
- Start early and sustain efforts post "go-live."
- Constant communication of project benefits.
- Integrate change activities into the project plan.
- Budget resources specifically for change activities.
- Tactics include top management support, user involvement, training, support channels, and external consultants.

When it comes to managing and leading a team/project, eventually you are managing what? - People and personal relationships.